EXP NO:16 Experiment to understand EDA-Quantitative and Qualitative analysis.

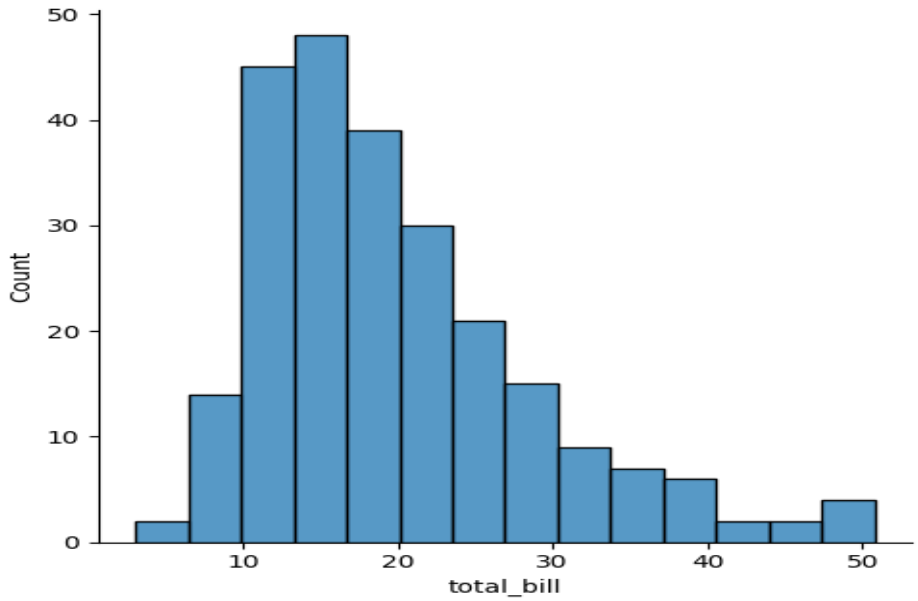
AIM:

To make an experiment to understand EDA-Quantitative and Qualitative analysis.

PROGRAM:

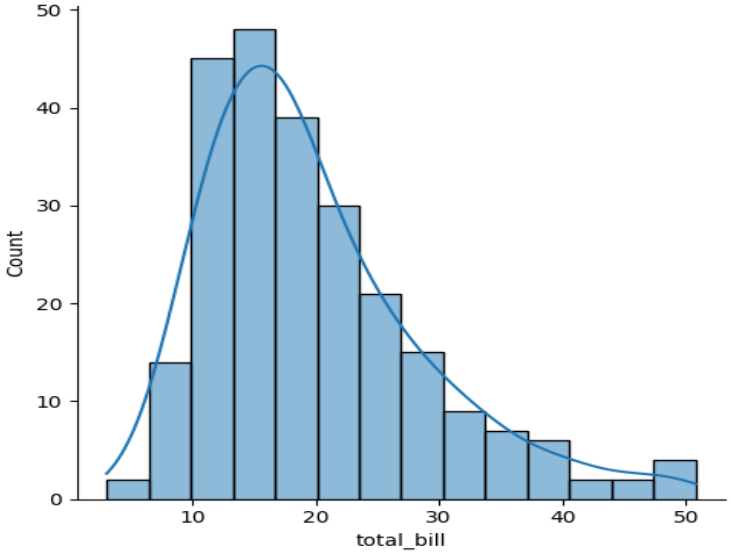
```
 import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
tips = sns.load_dataset('tips')
print(tips.head())
sns.displot(data=tips, x='total_bill', kde=True)
plt.show()
sns.displot(data=tips, x='total_bill', kde=False)
plt.show()
sns.jointplot(data=tips, x='tip', y='total_bill')
plt.show()
sns.jointplot(data=tips, x='tip', y='total_bill', kind='reg')
plt.show()
sns.jointplot(data=tips, x='tip', y='total_bill', kind='hex')
plt.show()
sns.pairplot(tips)
plt.show()
print(tips.time.value_counts())
sns.pairplot(tips, hue='time')
plt.show()
sns.pairplot(tips, hue='day')
plt.show()
plt.figure(figsize=(8, 6))
sns.heatmap(tips.corr(numeric_only=True), annot=True)
plt.show()
plt.figure(figsize=(8, 5))
sns.boxplot(x=tips.total_bill)
plt.show()
plt.figure(figsize=(8, 5))
sns.boxplot(x=tips.tip)
plt.show()
plt.figure(figsize=(8, 5))
sns.countplot(data=tips, x='day')
plt.show()
plt.figure(figsize=(8, 5))
sns.countplot(data=tips, x='sex')
plt.show()

tips.sex.value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.show()
tips.sex.value_counts().plot(kind='bar')
```



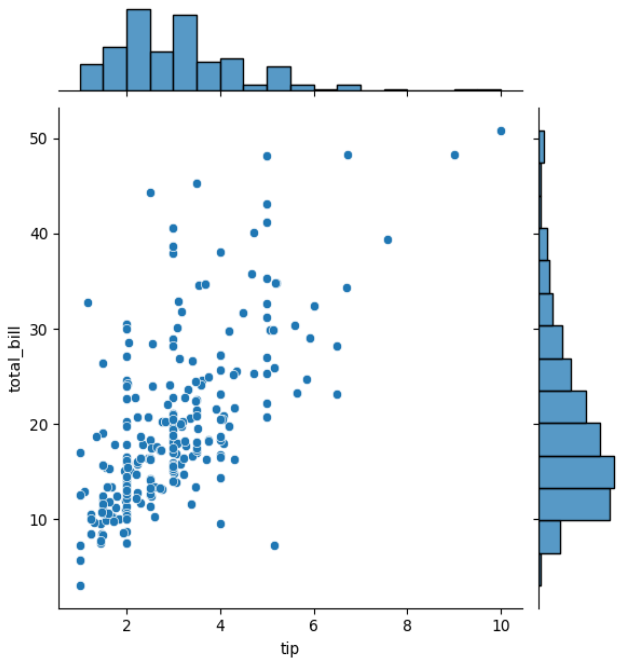
plt.show()

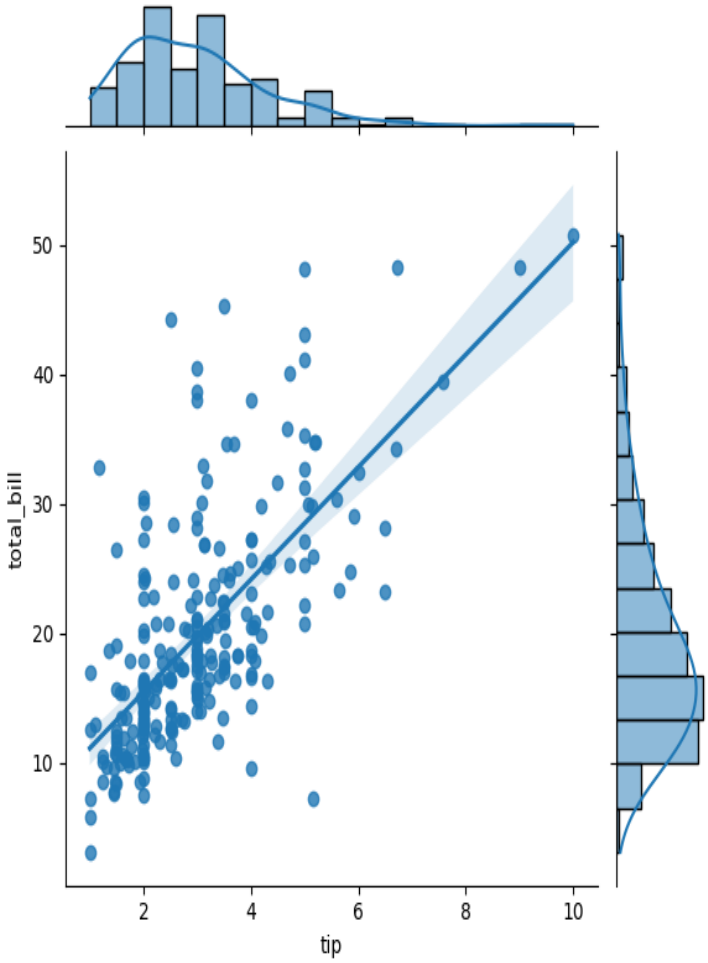
```
plt.figure(figsize=(8, 5))
sns.countplot(data=tips[tips.time == 'Dinner'], x='day')
plt.show()
```

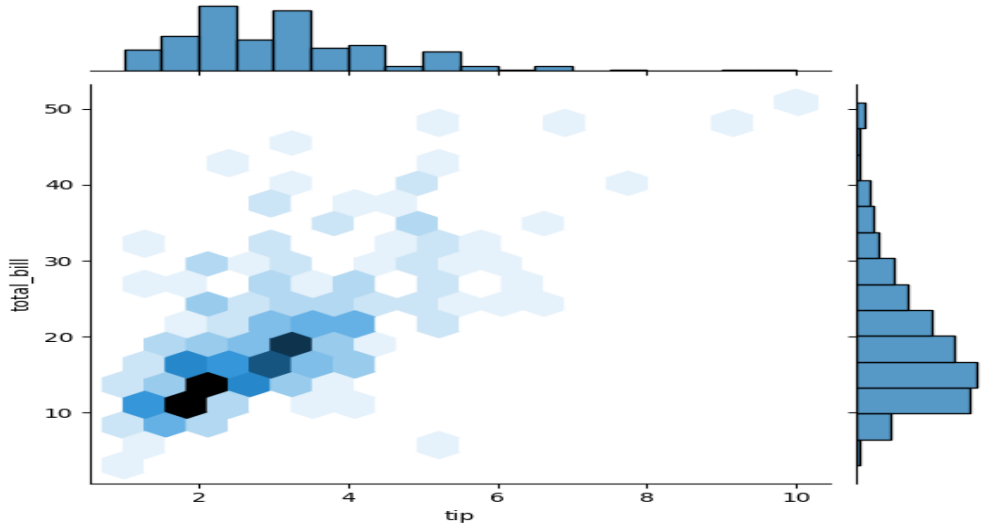


OUTPUT:

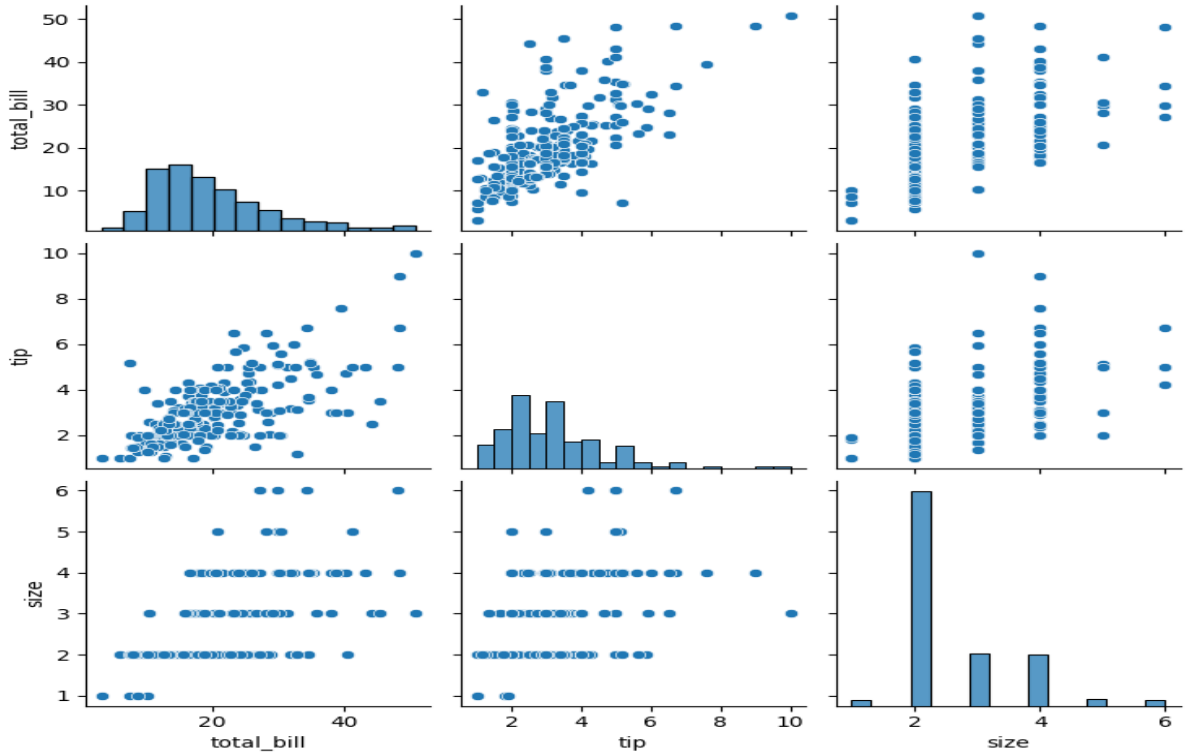
	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

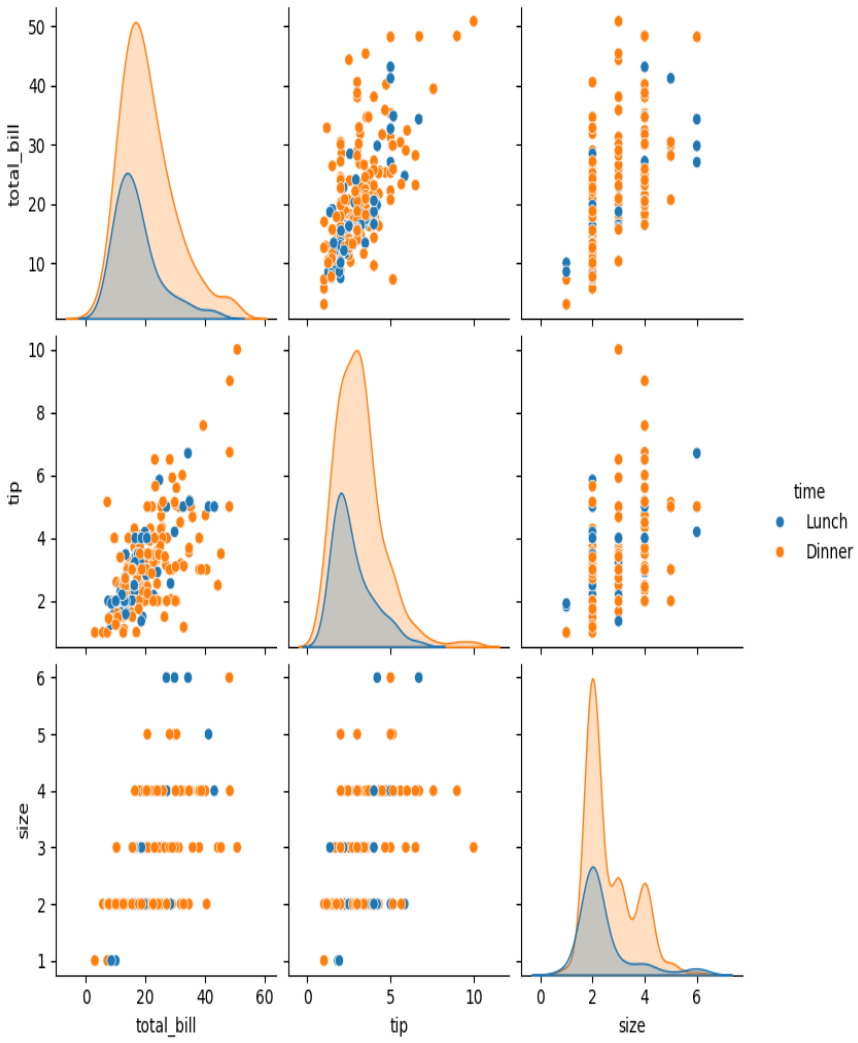

total_bill
tip
50
40
30
20
10
2
4
6
8
10

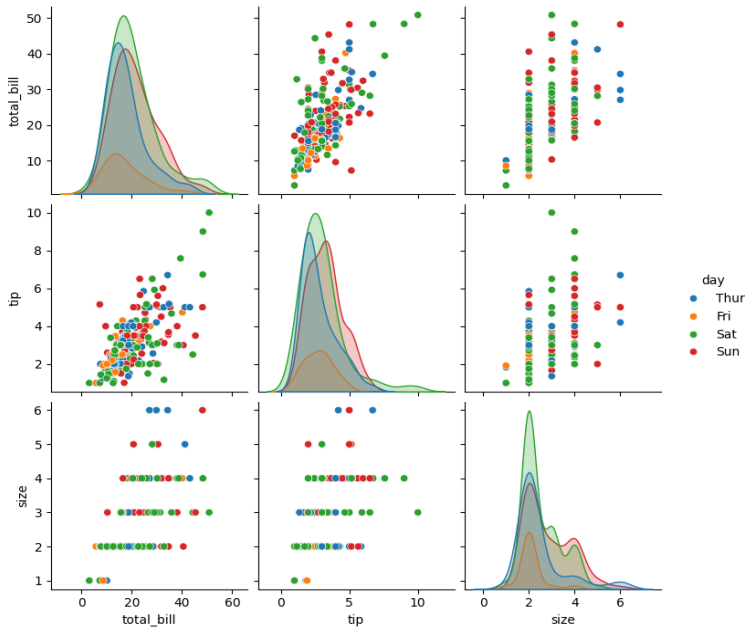

total_bill
tip
50
40
30
20
10
2
4
6
8
10

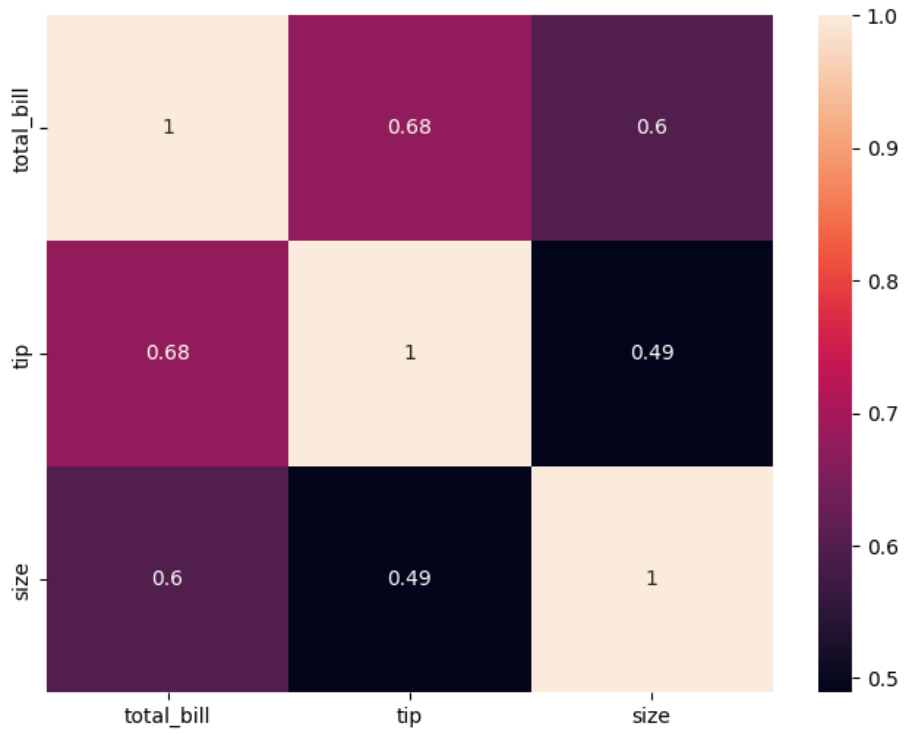


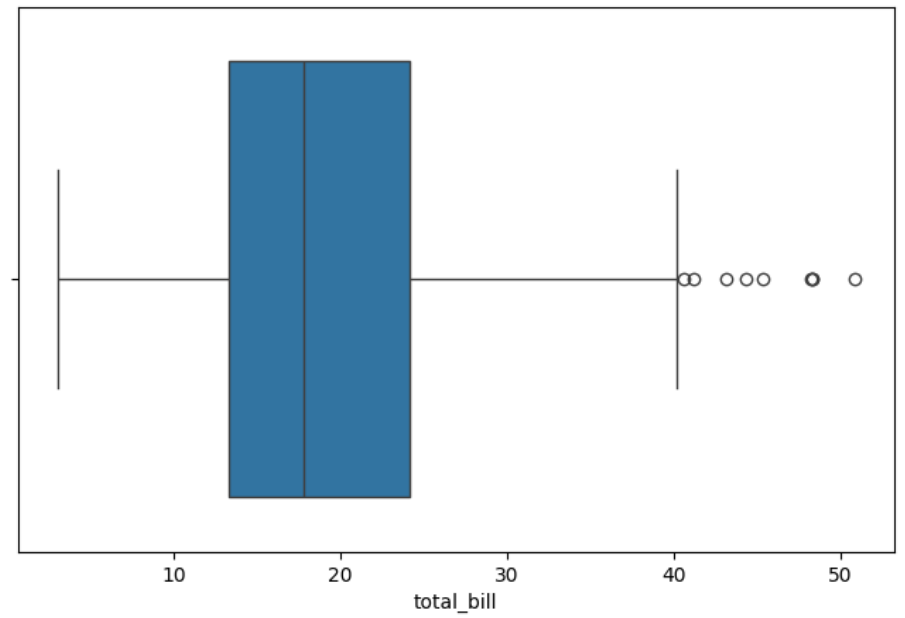
time
Dinner 176
Lunch 68
Name: count, dtype: int64

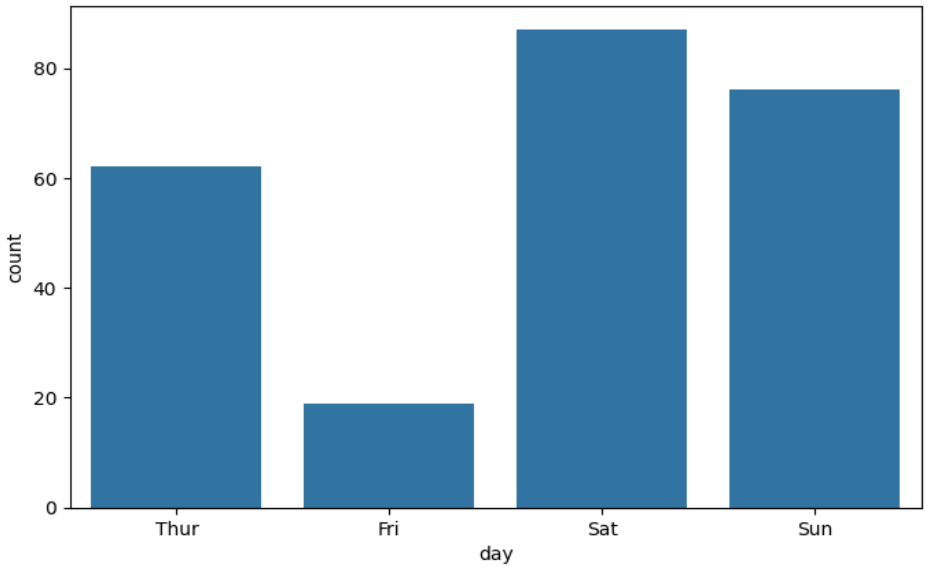


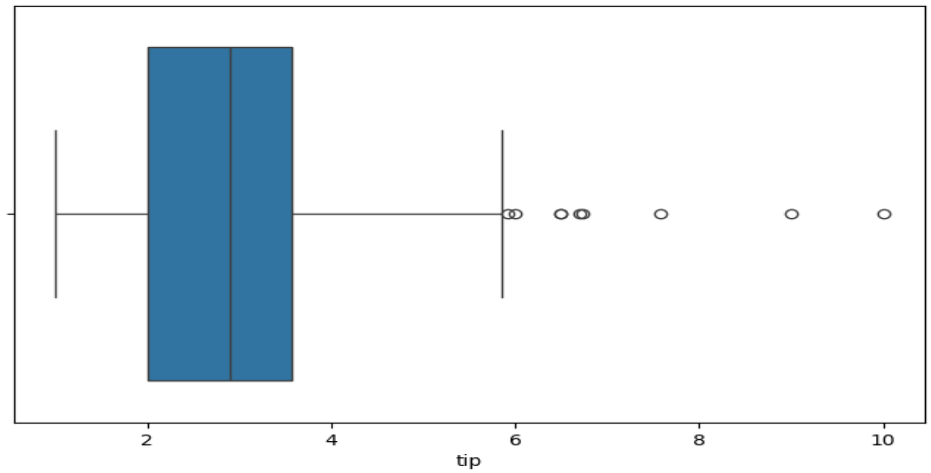

total_bill
tip
size
time
Lunch
Dinner

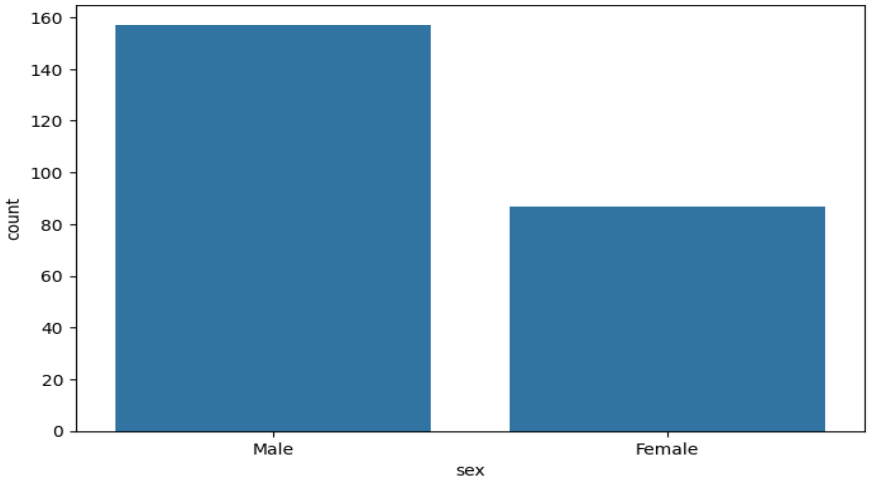

total_bill
tip
size
day
Thur
Fri
Sat
Sun

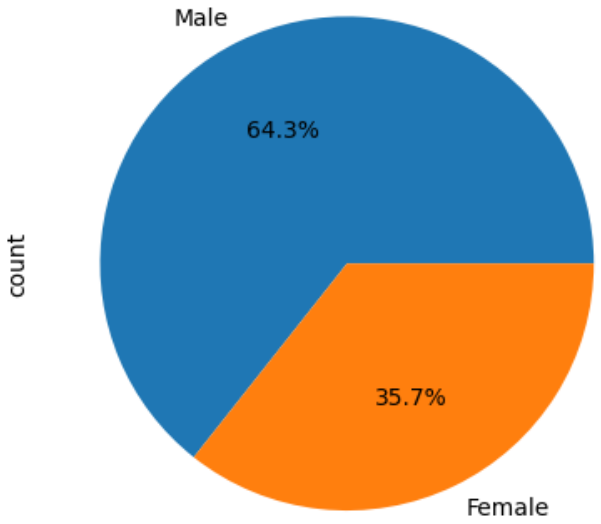

total_bill
tip
size
1
0.68
0.6
0.68
1
0.49
0.6
0.49
1
total_bill
tip
size
1.0
0.9
0.8
0.7
0.6
0.5

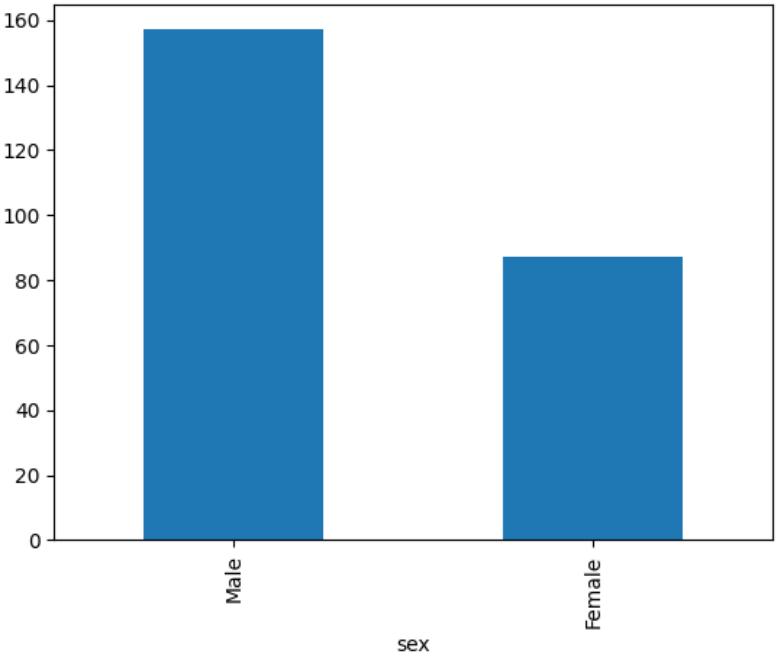

10
20
30
40
50
total_bill

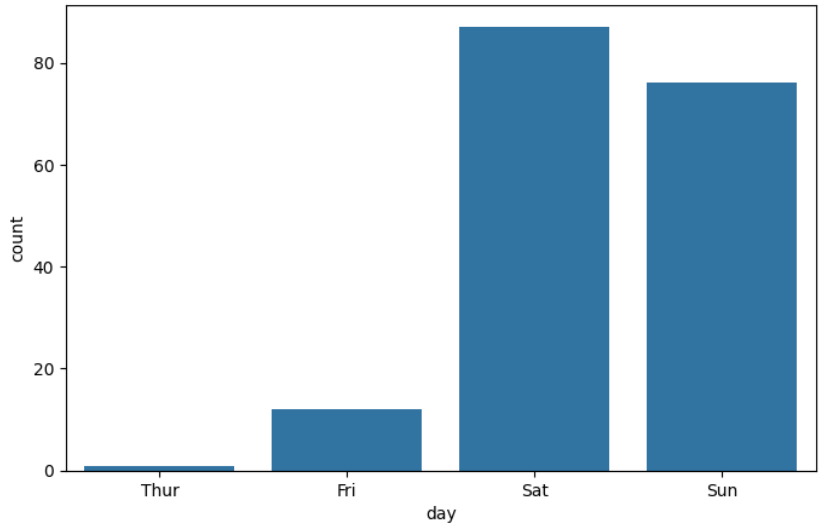

80
60
40
20
0
count
Thur
Fri
Sat
Sun
day


2
4
6
8
10
tip


160
140
120
100
80
60
40
20
0
count
Male
Female
sex


Male
64.3%
count
35.7%
Female


160
140
120
100
80
60
40
20
0
Male
Female
sex



RESULT:

Thus the experiment to understand EDA-Quantitative and Qualitative analysis is done.